What we're doing

RL on #230's `<contacts-v1.multi>` checkpoint, with the reward computed over the **sections of a single rollout** rather than over the rollouts of a group.

Four arms, all warm-started from `plm-exp230-cv1-multi-1_5b-.../step-1988`: **M-C** rewards each section by its leave-one-out contribution to its own rollout's consensus; **M-F** rewards the last section's F1; **M-B** rewards the best section's F1 (an oracle ceiling); **M-0** is a zero-LR control.

Why

#208 ran eleven scored runs across five reward designs and none improved consensus R-precision. The mechanism was a **unit mismatch**: the reward acted on one rollout, while the metric scored a vote over 100 *independent* rollouts — an object no single rollout can see. Making each rollout individually better made the hundred redundant, and consensus scoring cannot rank a pair that no rollout emits.

Under `<contacts-v1.multi>` the candidate set lives **inside one rollout** — ~22 contact sets in one sequence. A reward on the aggregate of one rollout's sections is therefore computed on the same kind of object the metric scores, and its credit assignment is within the sequence, where the gradient can reach it.

That is the only reason to expect a different outcome from #208, and it is the one thing this experiment tests.

The risk, preregistered

#230's checkpoint already sits at mean pairwise Jaccard **0.304** between its sections, past exp200's 0.30 diversity-collapse criterion **before any RL**. #208's dominant failure was RL collapsing diversity. So union coverage, total votes and votes-per-pair are reported every batch, and three diversity gates are kill criteria rather than diagnostics.

Results — the hypothesis is half right

Moving the reward's unit from *rollouts of a group* to *sections of one rollout* produces what #208 could not: an RL checkpoint that **improves consensus R-precision**. Arm M-C step-18 is the best on every aggregation mode — consensus 0.5750 (+0.0077), oracle-best 0.5578 (+0.0235), last 0.5267 (+0.0701). On eval2-natural, last goes 0.1696 → 0.2421.

It does not close the gap to independent sampling. 22 plain rollouts read 0.5896 against 0.5750 — the closest a multi-mode number has come, and 0.0146 short.

#208's result is a dose-response, not a verdict

Consensus against distance moved: 0.5673 at KL 0, **0.5750 at 0.007**, 0.5741 at 0.016, 0.5529 at 0.031, 0.3969 at 0.486. Every reward here helps at small KL and damages at large. #208 ran its arms at KL 0.06–0.10 and to 3.96 — past the peak on all of them — and its two arms under 0.0015 never moved at all. The window it needed lay **between the two learning rates it tried**.

Reward shape decides how a run fails, not whether

M-C and M-F halve the contacts emitted while becoming *more* diverse (Jaccard 0.60× and 0.44×). M-B holds the volume and emits the same contacts 1.7× as often. Both routes end at the same coverage floor. #208 found these two modes across different reward families; here they are produced deliberately by reward shape, on one model and one data order.

Why M-C collapsed: the reward is not scale-free

`m_k = C(all) − C(all \ {k})` grows as a rollout emits *fewer* sections, because each survivor is then more load-bearing. So a rollout can raise the reward on every one of its sections by making its own consensus worse.

Measured by truncating real rollouts: the reward per section is **366× larger at 1 section than at 22**, while the rollout's own consensus falls 0.543 → 0.341. In groups differing in nothing but section count, a one-section rollout gets **+4.80** advantage and a 22-section rollout **−0.22**.

`E[A] = 0` holds exactly throughout — centring is computed *within* the quantity being gamed. This explains the accelerating collapse (13.7 → 1.1 sections over five steps) and why M-F and M-B, which have no such term, kept or grew theirs.

The clause for #208's rule: `E[r] = 0` constrains the mean over the candidates the policy emitted; it says nothing about whether the reward's *scale* depends on how many that was.

The other lesson: gate on `union/R`, and on precision

The preregistered coverage gate stopped all three arms; union/R never left 2.8–4.6 in any of them, including the collapsed one, whose union/R was *higher* than the warm start's. Coverage was never binding. Precision (0.50 → 0.14) was.

Figures

`plots/dose_response.png` — R-precision against distance moved, three aggregation modes, with the budget-matched bar drawn on.

`plots/gates_over_training.png` — the diversity gates per batch, rolling median.

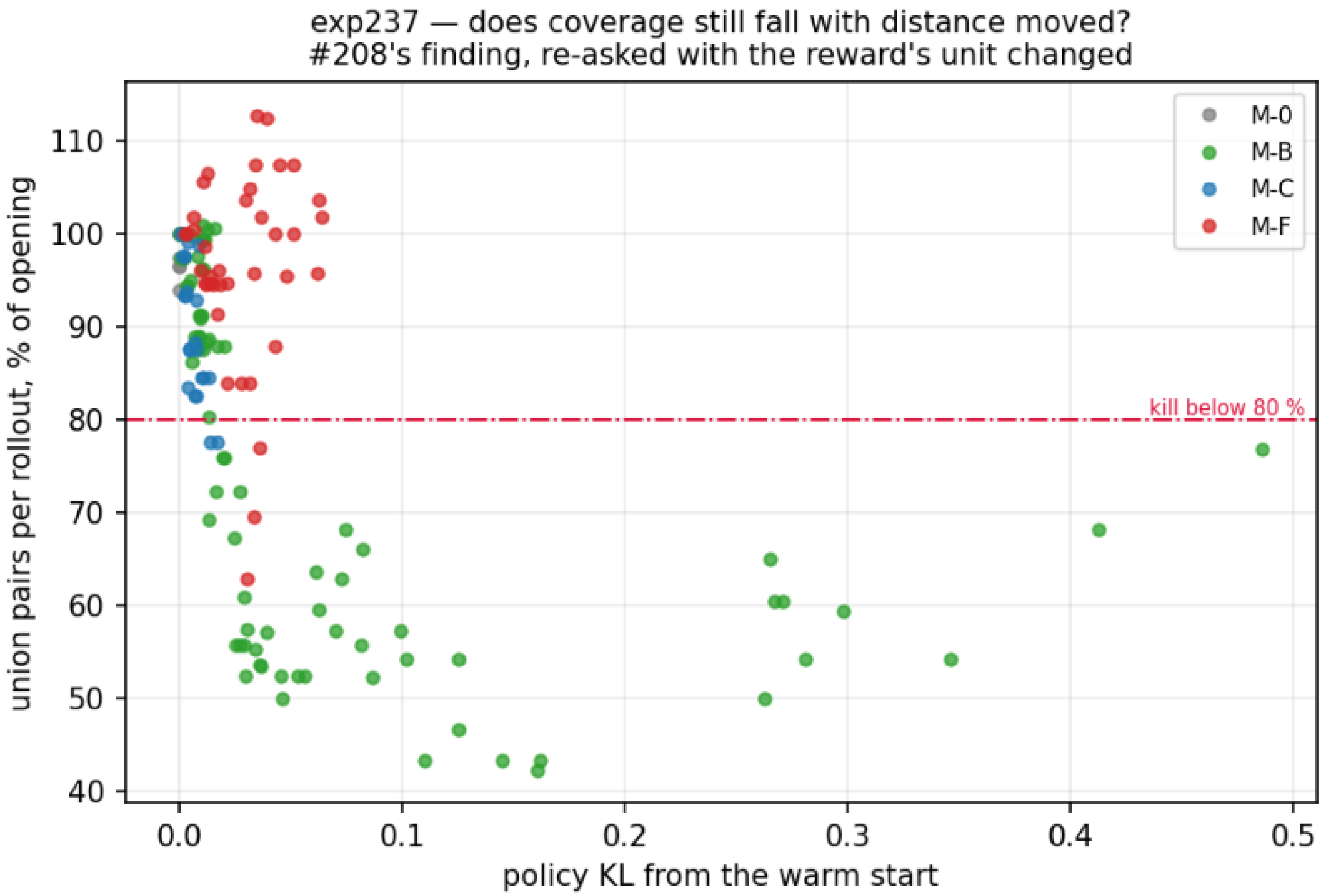
`plots/reward_shape.png` — arm M-C's marginal distribution and its group-centred advantage, i.e. the mechanism this run proposed and then refuted.

`plots/section_count_incentive.png` — the scale pathology: 366x the reward for a worse answer.

`plots/coverage_vs_kl.png` — coverage against distance, #208's question re-asked.

coverage_vs_kl.png

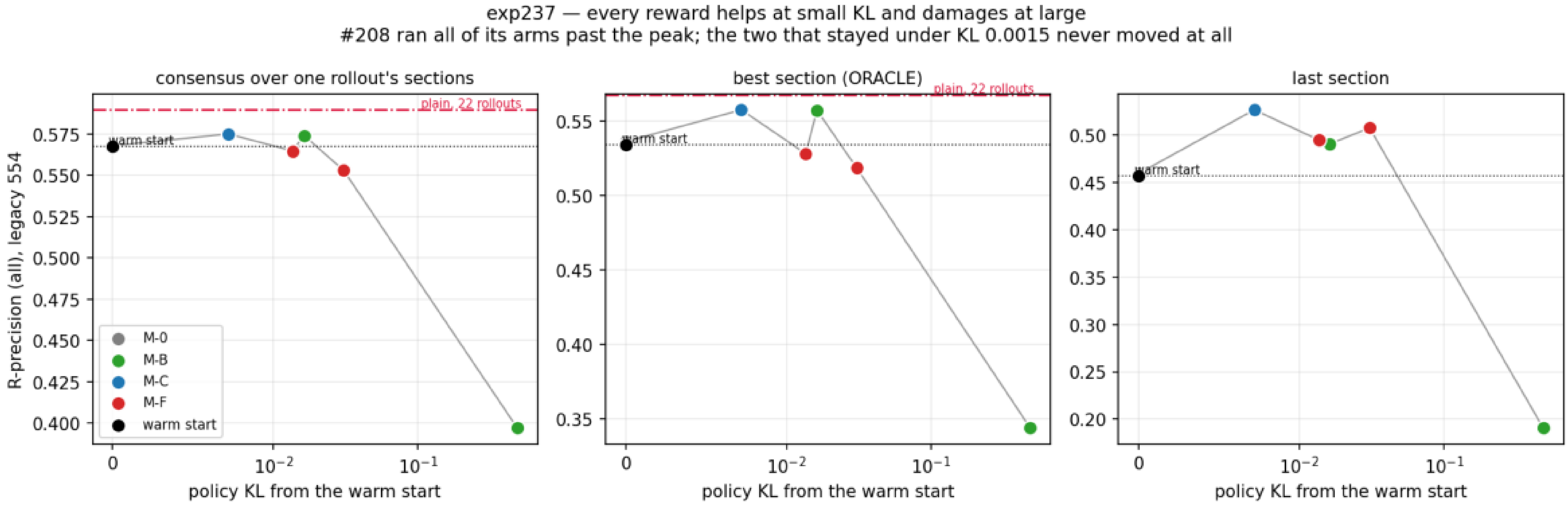
Union coverage against distance moved. M-C crosses the 80% floor by KL 0.013, M-B by 0.016, M-F not until 0.036 -- same destination, different cost per unit of KL.



`$ make_plots.py --steps data/training_steps.csv.gz --out plots/`

dose_response.png

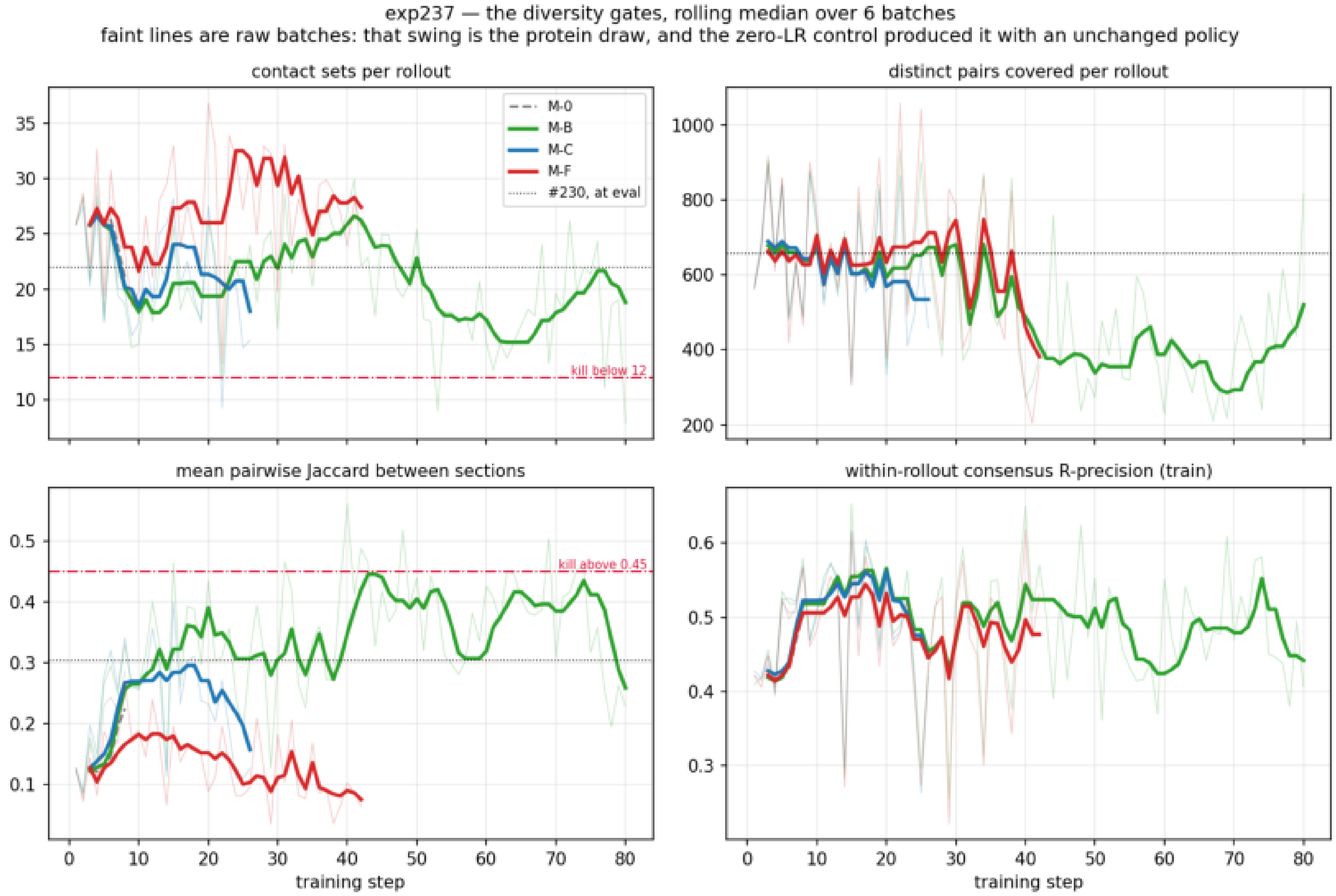
R-precision against distance moved. Consensus peaks at KL 0.007-0.016; #208 ran every arm past it. Red = the budget-matched bar, which nothing reaches.



$ plot_reward_shape.py --data data --out plots

gates_over_training.png

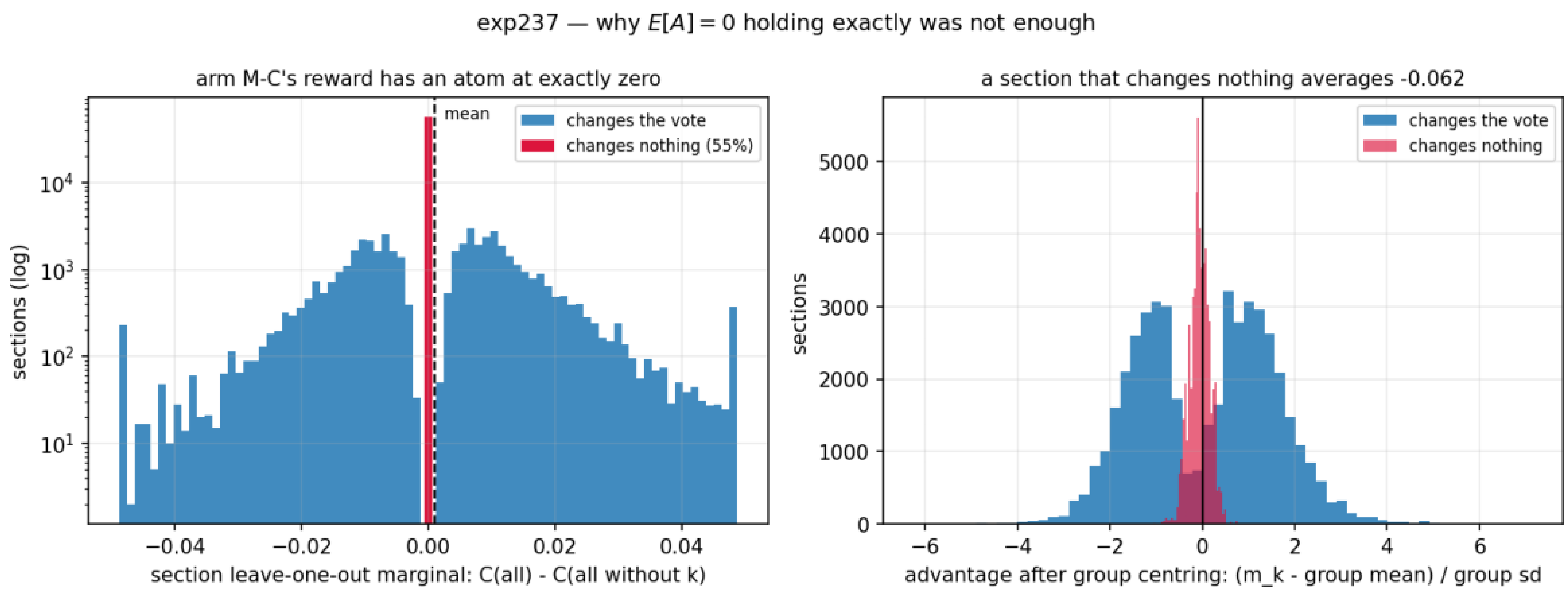
The diversity gates per batch, rolling median over 6. Jaccard separates the arms: M-B rises, M-C and M-F fall. Faint = raw batches, i.e. the protein draw.



`$ make_plots.py --steps data/training_steps.csv.gz --out plots/`

reward_shape.png

Arm M-C's reward before training: 55% of sections change no vote, and average -0.062 once centred. M-F has no such atom and collapsed the same -- so this is not the cause. See RESULTS.md.

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$ plot_reward_shape.py --data data --out plots
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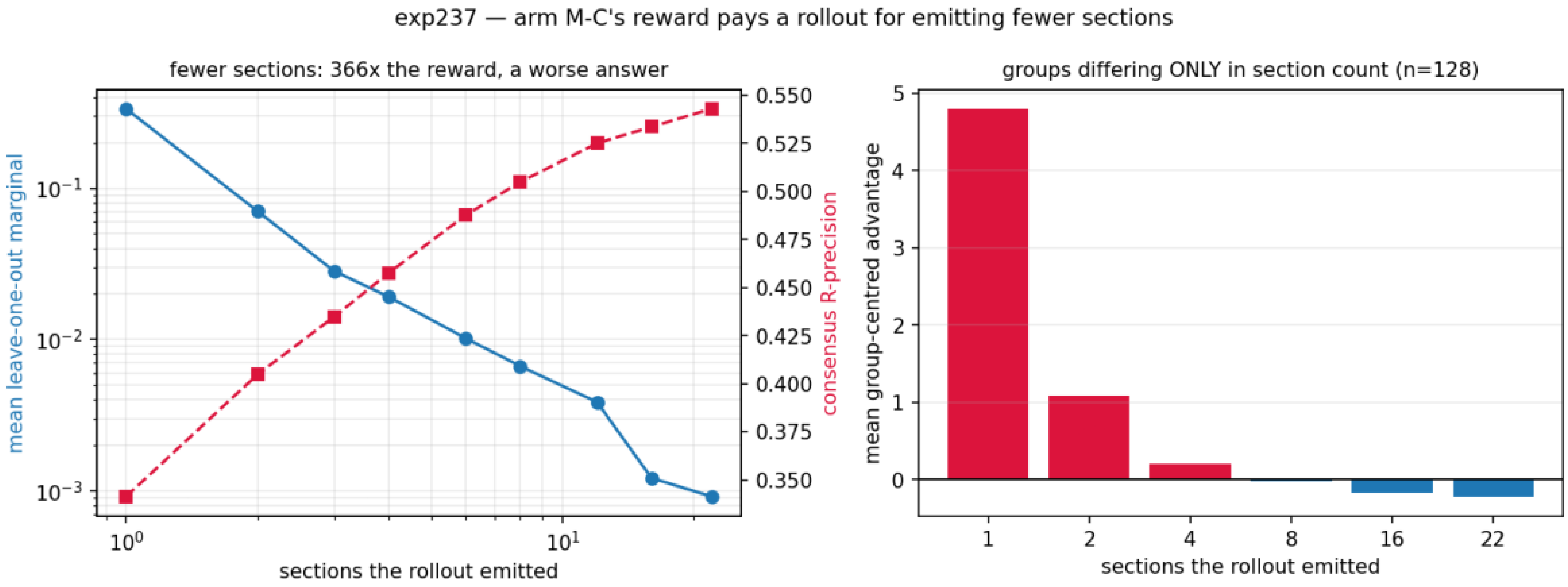
section_count_incentive.png

m_k is not scale-free in the section count: at 1 section it is 366x its value at 22, while the rollout's own consensus falls 0.543 -> 0.341. Centring does not remove it -- the shorter rollout is the one above its group's mean.



$ plot_reward_shape.py --data data --out plots